Lecture Notes:
Math & Statistics for Data Science
Unit 4 – Prob Stat

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1 Random Variables

Learning Goals

- Define random variables and distinguish them from constants.
- Understand experiments, outcomes, sample spaces, and events.
- Differentiate between discrete and continuous random variables.

Content

- **Motivation:**
  - Constants (e.g., whiteboard length) vs. random variables (stock price, temperature, die rolls).
  - Random does not mean "no pattern" — it means the pattern is too complex to predict exactly.
- **Definitions:**
  - Experiment = measurement process.
  - Outcome = result of a single experiment.
  - Random variable = mathematical representation of outcome.
  - Sample space (S) = set of all possible outcomes.

- Event (E) = subset of the sample space.
- Probability $P(E)$: maps events to numbers in $[0, 1]$.

- **Types of random variables:**
  - Discrete/finite: rolling a die.
  - Discrete/infinite: choosing a random integer.
  - Continuous: temperature.

- **Notation:**
  - Random variables: capital letters (e.g., X).
  - Values: lowercase (e.g., x).
  - Example: $P(X = 6)$ = probability that die lands on 6.

Exercises

- Identify sample spaces for coin flips, dice, and continuous measurements.
- Classify variables as discrete or continuous.

2 Probability Rules

Learning Goals

- Apply probability laws with complements, unions, and intersections.
- Use conditional probability and independence.

Content

- **Set operations:** complement, union, intersection.
- **Conditional probability:**
  - Definition: $P(x|y)$.

  - Multiplication rule:

$$P(x \wedge y) = P(x|y)P(y) = P(y|x)P(x).$$

- **Independence:**
  - If $P(x|y) = P(x)$, then x and y are independent.
- **Examples:** overlapping vs. non-overlapping student classes.

Exercises

- Compute conditional probabilities in classroom-type scenarios.
- Test whether given events are independent.

3 Sampling and Distributions

Learning Goals

- Explain the role of sampling in probability/statistics.
- Distinguish between discrete and continuous distributions.
- Use expectation values and understand the law of large numbers.

Content

- **Sampling:** experiments produce observed values of random variables.
- **Distributions:** describe probability across all outcomes.
- **Discrete distributions:**
  - Uniform distribution: coin flips, dice ($P(X) = 1/n$).
  - Geometric distribution.
- **Continuous distributions:**
  - Examples: height, IQ, reaction time, stock returns.

  - Normal distribution: bell-shaped, centered at mean.

- **Expectation values:**
  - Population mean vs. sample mean.
  - Law of large numbers: as sample size grows, sample mean $\rightarrow$ true mean.

- **Applications: Monte Carlo methods.**
  - Estimate π.
  - Approximate integrals.

Exercises

- Simulate coin flips/dice rolls and compare to theoretical probabilities.
- Compute expected values of simple random variables.
- Use Monte Carlo simulation in Python to approximate π.

Pedagogical Notes

- Emphasize intuition: randomness means complexity, not chaos.
- Use real-world data (e.g., stock prices, temperatures) for continuous distributions.
- Incorporate Python tools for simulation, sampling, and visualization.
- Highlight connections to data science: Monte Carlo, inference, model evaluation.